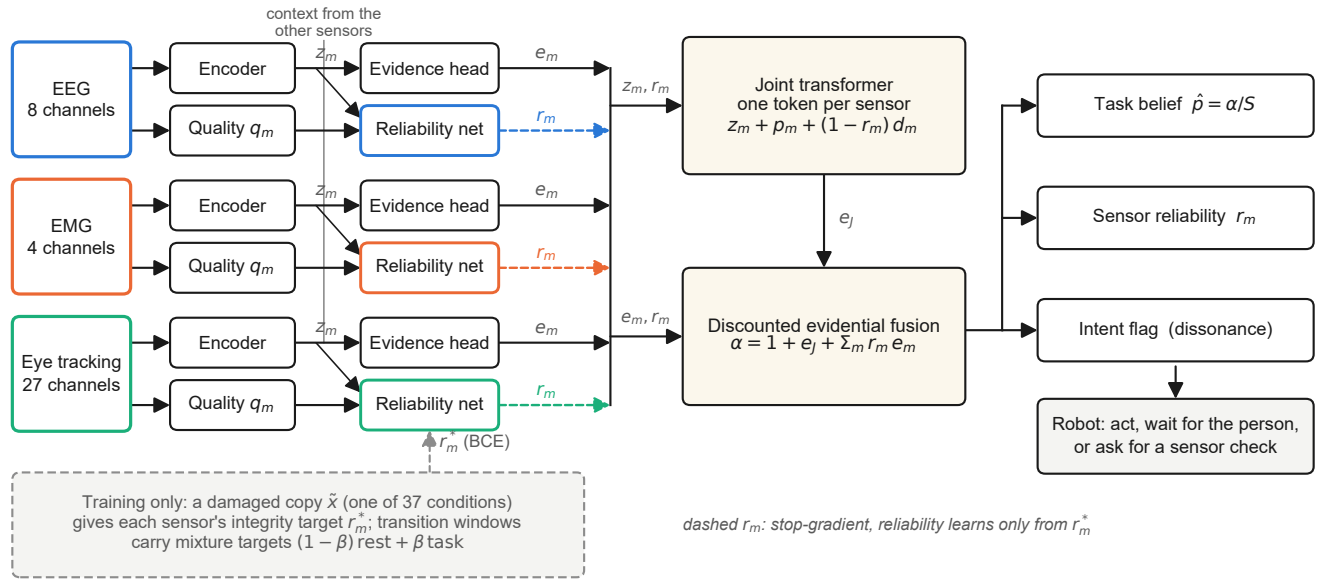

context from the other sensors
EEG 8 channels
EMG 4 channels
Eye tracking 27 channels
Encoder
Quality q_m
Evidence head
Reliability net
z_m
e_m
r_m
z_m, r_m
e_m, r_m
Joint transformer one token per sensor $z_m + p_m + (1 - r_m)\, d_m$
e_J
Discounted evidential fusion $\alpha = 1 + e_J + \Sigma_m\, r_m\, e_m$
Task belief $\hat{p} = \alpha / S$
Sensor reliability r_m
Intent flag (dissonance)
Robot: act, wait for the person, or ask for a sensor check
r_m^* (BCE)
Training only: a damaged copy $\tilde{x}$ (one of 37 conditions) gives each sensor's integrity target r_m^*; transition windows carry mixture targets $(1 - \beta)$ rest $+ \beta$ task
dashed r_m: stop-gradient, reliability learns only from r_m^*